

First name:

Last name:

Student ID:

SOLUTIONS

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Problem 1. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

For spherical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

[5 marks]

Solution.

$$\text{As } \phi = \text{const} = \pi/6, \quad dA = (h_\rho d\rho)(h_\theta d\theta)$$

$$\Rightarrow dA = \rho \sin \phi d\rho d\theta = \rho \sin(\pi/6) d\rho d\theta$$

$$\Rightarrow dA = \frac{1}{2} \rho d\rho d\theta$$

Then area of cone

$$= \int_0^{2\pi} \int_0^2 \frac{1}{2} \rho d\rho d\theta = \frac{1}{2} \int_0^{2\pi} d\theta \int_0^2 \rho d\rho$$

$$= \frac{1}{2} (2\pi) \left(\frac{\rho^2}{2} \Big|_0^2 \right) = \pi \left(\frac{2^2}{2} - \frac{0^2}{2} \right)$$

$$= 2\pi$$

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Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

[5 marks]

Solution.

$$\begin{aligned} \text{As } r = \text{const} = 2, \quad dA &= (h_\theta d\theta)(h_z dz) \\ &= r d\theta dz = 2 d\theta dz \end{aligned}$$

Then surface area:

$$\begin{aligned} \int_0^{2\pi} \int_0^2 2 dz d\theta &= 2 \int_0^{2\pi} d\theta \int_0^2 dz \\ &= 2(2\pi)(2-0) = 8\pi. \end{aligned}$$

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